

tive measures, their benefit depends on the type of data and the degree of redundancy or compressibility. Thin provisioning, however, can provide benefit regardless of the type of data," says Joshi.

Symantec recently launched FileStore, a software solution that enables organisations to build scalable, high-performance, file-based storage services for their enterprise, including private and public clouds.

IBM's TotalStorage portfolio of solutions also offers options to consolidate storage that can scale up with large-capacity storage products, or scale out with rack-and-stack products that grow modularly.

Storage virtualisation

Virtualisation is another trend that is almost inseparable from enterprise storage now. "Storage virtualisation and consolidation brings forth unique advantages of realising better asset utilisation along with better availability. Virtualisation platforms provide much desired flexibility into the infrastructure," says Dutta of IBM.

The IBM System Storage SAN Volume Controller (SVC) is a storage virtualisation system that enables a single point of control for storage resources to help support improved business application availability and greater resource utilisation.

Another of IBM's virtualisation-related product is the IBM Information Infrastructure – the infrastructure management platform that focuses on storage monitoring, including other infrastructure elements such as tapes, network fabric and file systems within its purview.

Cloud computing

Cloud computing, storage offered as a service, is literally levitating organisations to cloud nine as large-scale storage is now within the reach of an organisation which does not have to worry about maintenance. An organisation can now 'rent' storage space and use it.



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Joshi of Symantec says, "By using cloud computing, organisations can tap cost savings over traditional self-hosting of infrastructure and software, enhance system performance, and increase the agility in delivering their products and services to their own customers."

"Clearly, we are seeing a lot of drive in cloud computing—software, storage, platforms—practically all the aspects of information management can be offered as a service through clouds," says Gupta of EMC.

The recent launches of most industry majors are centred around cloud computing.

"This complements our unified storage offering by ensuring 'unification of scaling' along with connectivity architectures like SAN, NAS and iSCSI. As this phased approach moves forward, customers using our cloud com-

puting platform Data ONTAP 8 will be able to leverage the combined scale-up and scale-out capabilities along with new technologies and features that further enable the cloud," says Syed Masroor, director-systems engineering, NetApp India.

"Virtualisation, Fibre Channel over Ethernet and tiered-storage are the key technological changes that have happened in the last few years and are now extensively being accepted by customers," he adds.

Virtual data centres

Till now we have been storing information in physical data centres. But sooner or later, we will reach a limit on the physical capacities of data centres. This, according to Gupta, explains why virtual data centres (VDC) are so important today.

EMC, Cisco and VMware have a joint vision for virtual data centres and have jointly created exciting products like the Symmetrix V-Max, which promises to change the industry perception towards storage.

V-Max represents a massive redesign of architecture and operating environment. EMC claims that it is the first high-end array built on a scale-up and scale-out architecture, as well as the first high-end array with industry standard components.

Disaster recovery

Data deduplication, virtualisation, advanced networking technologies, have all made backups easier and more effective at the enterprise level. Consequently, advances in snapshot and backup technology have improved recovery point objectives, recovery time objectives and disaster recovery.

Recently, Tandberg Data launched the DPS 1000 series of virtual tape library (VTL) products. The DPS1200 VTL creates a unique and secure virtual library connection for each host system, including laptops, desktops, workstations, servers and virtual machines. "This isolation of data helps to satisfy regulatory compliance concern-